BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 9231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** 9231

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. Main events take place during metaphase-II are:

(A) Chromosome become visible

(B) Nucleoli and nuclear envelope disappear

(C) Chromosomes arrange themselves on equatorial plate

(D) Nucleoli and nuclear membrane reappear

Q2. Enzyme cellulose speed up the:

(A) Digestion of protein only

(B) Digestion of starch only

(C) Digestion of lipids only

(D) Digestion of cellulose only

Q3. In light reaction, oxygen is released while, hydrogen atoms are used to reduce:

(A) ADP to Pi

(B) H_2O to O_2

(C) ADP to ATP

(D) NADP to NADPH

Q4. Rate of photosynthesis is affected by very intense light because it damages:

(A) Chromoplast

(B) Leucoplast

(C) Chlorophyll

(D) Root

Q5. The common symptoms of diarrhoea are:

(A) Abdominal burning and pain

(B) Loss of appetite and weight loss

(C) Loose bowel movements and vomiting

(D) Dry stool and abdominal bloating

Q6. The function of the pericardial fluid is to:

(A) Protect the heart
(B) Lubricate the heart
(C) Protect the lungs
(D) Lubricate the lungs

Q7. The use of organisms for the welfare of humanity is studied in:

(A) Microbiology
(B) Parasitology
(C) Socio-Biology
(D) Biotechnology

Q8. In scientific method, biologists conclude results from:

(A) Observations
(B) Experiments
(C) Hypotheses
(D) Deductions

Q9. Mushrooms look like plants but they are not plants because they lack:

(A) Roots
(B) Seeds
(C) Flowers
(D) Chloroplast

Q10. In animals, muscle undergo contraction and play role in:

(A) Respiration of body
(B) Growth of brain
(C) Movement of body
(D) Circulation of blood

Q11. Eukaryotic ribosomes are slightly larger than prokaryotic ribosomes. But they are involved in:

(A) Respiration of body
(B) Protein synthesis
(C) Movement of body
(D) Bone development

Q12. Mitosis plays an important role in:

(A) Gametes formation
(B) Genetic variation
(C) Regeneration
(D) Fertilization

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32 | **Paper Code:** 9231

Q1. Attempt any eight of the following short questions. Each question carries 4 marks.

(i) How biologist can adopt agriculturist as a career? Give FOUR reasons.

(ii) Explain the living characteristics of virus.

(iii) Write contribution of Schleiden and Schwann in cell biology.

(iv) Briefly describe the structure of ground tissues in plants.

(v) Define interphase stage of cell cycle. Also write names of any TWO events of interphase.

(vi) How cytokinesis of animal cell is different from plant cell?

(vii) Write 4 steps of enzyme action.

(viii) Following equation is of Aerobic Respiration. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 36ATP$ (net gain) Answer the following questions. (i) Write names of reactants given in above equation. (ii) Write names of two products in which organic food is completely breakdown. (iii) Write complete name of ATP.

(ix) Distinguish between carbohydrates and proteins with the help of following table. Table headings: Properties | Carbohydrates | Proteins. Rows: Main sources; Energy value.

(x) How do plants control transpiration?

(xi) Differentiate between pulmonary circulation and systematic circulation.

SECTION-C

Maximum Marks: 21

> Note: Attempt any three of the following questions. Each question carries 7 marks.

Q2.

(i) Differentiate between ratio and proportion.

(ii) Why conservation of biodiversity is important? List FOUR reasons.

Q3.

(i) Highlight the differences between ribosome and cell wall of eukaryotic cell and prokaryotic cell.

(ii) Define mitotic apparatus in animal cells.

Q4.

(i) Justify that enzymes are specific in action.

(ii) Explain glycolysis as an important step of aerobic respiration.

Q5.

(i) Write down any THREE advantages of organic fertilizers.

(ii) Give any TWO differences between artery and vein with reference to their functions.